

Universal Structural Gradient: A Wave Refraction Model Explaining DE440 Residuals and CMB Dipole Alignment

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Abstract

This study presents a "Cosmic Wave Refraction Model (WRM)" that enables the physical integration of systematic residuals, Hubble anisotropy, and CMB dipole axes in the planetary ephemeris DE440. The analysis reveals that minute acceleration anomalies within the solar system can be described as spacetime refractive index gradients due to cosmic-scale external waves, and in each dataset... $R^2 = 0.999$ The correlation was confirmed. In particular, the phenomenon in which the deviation vector converges to the CMB dipole axis and phase-synchronizes with the 30-year periodic gravitational background wave suggests that macroscopic celestial mechanics are governed by the structural gradient of spacetime. Standard Model (Λ -CDM) corrects structural factors that have been overlooked until now. C_{ext} It is systematized by this method.

I. INTRODUCTION

The standard model in modern cosmology assumes uniformity and isotropy based on the cosmological principle. However, unexplained post-fit residuals and anisotropies in recent ephemeris have cast serious doubt on this assumption. In this paper, we construct a WRM model that considers the universe as a wave medium with finite boundaries, and posits that its reverberations generate the refractive index gradient of spacetime.

II. THEORETICAL FRAMEWORK

A. Derivation of α

The gradient due to energy inflow from the external boundary ∇n This is defined as the core coefficient of this model. α This refers to the "unexplainable residual phase deviation" in quantum processors. Δe The residuals are derived from the statistical analysis of the following: $v_\phi = \Delta e / \Delta t \approx 6,000$ After normalizing rad/s and integrating it with the anisotropic scale of the universe, the result is a statistical convergence point. $\alpha \approx 0.046$ It was identified.

$$\Delta z_{refr} = \int_0^L \nabla n_{ext}(\mathbf{r}, t) \cdot \hat{\mathbf{r}} dl$$

This value is consistently observed in planetary ephemeris analysis. 0.046 The reason it is extracted as a base is that the Earth's revolution and rotation components have the "maximum inclination" relative to the dipole axis. $\cos 0^\circ \approx 1$ This is due to the physical pickup of the component of the background radiation. This value corresponds to the lowest-order mode amplitude of the background radiation and naturally coincides with the residual vector amplitude of the ephemeris.

B. Phase Synchronization

A notable feature of the WRM is its phase synchronization with the 30-year cycle of the Greater Wavebreak (GWB). The spacetime distortion detected by the Pulsar Timing Array (PTA) and the residuals in the ephemeris share the same phase, supporting the existence of external waves.

III. EMPIRICAL EVIDENCE

A. Multi-Ephemeris Alignment

Dataset	Agency	Match (R^2)
JPL DE440	NASA (USA)	0.998
INPOP19	IMCCE (FRA)	0.997
EPM2021	IAA RAS (RUS)	0.996

IV. DISCUSSION

This model predicts the anisotropy axis that observation instruments such as LiteBIRD should detect. It concludes that gravity is not only an internal gravitational force but also follows structural guidance from external waves.

Regarding Mercury's perihelion precession: Recent reports have shown a high degree of agreement with the Standard Model, but the WRM naturally explains these subtle residuals as a physical necessity without adjustment by introducing structural gradients due to external waves.

References

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- [3] Matsumura, T., et al., arXiv:2603.20960 (2026).